CS5803 Data Visualisation Report

Title: Performance Insights in Valorant Champions Tour 2024: An Interactive Dashboard for Coach-Level Strategy

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1. Introduction

Dataset Description and Provenance

Dataset The data of this project comes from Valorant Champions Tour public 2024 match. It is the largest source of competitive esports stats within the world. The sanitised one, vct_combined_players. csv and consists of 3,123 rows and 16 columns documenting specifics on individuals' performance: 32 Columns at https://www.theopenfield.co/ Individual player statistics include average combat score (ACS), headshot percentage, kills per round, assists, agent play, and region. They are frequently employed in esports, among other contexts, for assessing players' skill, consistency and systematic game impact.

Another, vct_players_split_by_agent, is a variant to enable agent-specific analysis. csv by extracting the agents column and normalising to an agent per line. This change allowed for more accurate visualizations of agent distribution and role based performance. This dataset is collected from trust-worthy community-maintained repositories such as Kaggle and GitHub, hich collect stats on a frequent basis from official VCT broadcasts, match trackers, and Riot Games APIs. The size, format and dimensionality of the data are ideal for fine-grained visualization analysis with tools like Tableau and PowerBI.

Data Dictionary

Variable Name	Description
playerName	Name of the Valorant player
team	Player's affiliated professional team
region	Regional league affiliation (e.g., NA, EMEA, APAC)
rating	General rating of player's performance
average_combat_score	Average combat score (ACS)
average_damage_per_round	Average damage dealt per round (ADR)
kill_deaths	Kill to death ratio
kills_per_round	Average kills per round
assists_per_round	Average assists per round
headshot_percentage	Percentage of kills with headshots
clutch_success_percentage	Percentage of successful 1vX clutches
first_kills_per_round	First kills per round (entry fragging)

first_deaths_per_round	First deaths per round
agent	Character/role used by the player

User Persona

Name: Alex Kim

Position: Head Coach, Team Sentinels

Background: Alex is point on strategic planning and talent forward movement for Team Sentinels is an upper echelon esports org rampaging through the the Valorant Champions Tour. A veteran of competitive first person shooters for more than a decade, Alex has been breaking down and analyzing the CS:GO meta since it was in its infancy.

Objective: To find the highest performing players while investigating agent role trends, team and regional cognitive tendencies for team construction and roster construction in preparation for high-level play. Alex scouts opponents, monitors his roster and reports to team analysts using dashboards.

Planned Questions

Simple Questions:

- **Q1:** Who are the top 10 players based on Average Combat Score (ACS)?
- **Q2:** Which agents are most used by players with a rating greater than 1.1?
- **Q3:** How are high-rated players (rating > 1.1) distributed across regions and teams?

Complex Question:

- **CQ1:** How do ACS, headshot percentage, and damage per round interact across agents and regions?

2.Requirements Specification

Each of the visualisations was designed in order to answer one of the four research questions and were selected to represent the analytical requirements of a professional esports coach who surveyed performance data at a micro and macro level. The dashboard should allow to compare (between players), to categorize (per region, or per agent) and, to correlate (between different metric) informations. Visual encoding choices were based on effectiveness for comparison, ranking, part-to-whole, or distribution tasks.

R1: To respond to Q1, a horizontal bar plot by player Name of average_combat_score was applied. For ranking and comparing quantitative data, bar charts are appropriate. Horizontal writing makes it possible to read values, and especially names as in Fig. Applying team as color helps reinforce the members of the teams and allows for visual grouping across charts. Extended tool tips, including kill-death ratio and headshot rate, are shown to help interpret the ACS in more depth.

Treemaps do a great job at representing part-to-whole relationships and shine when it comes to categorical data. This design decision aligns with Few's (2009) recommendation to use treemaps for summarising hierarchical and categorical frequency data.

R3: Q3: A stacked column chart depicted how top-rated players are distributed across regions grouped by team. This information is depicted in the visualisation to allow for vertical (comparing across regions) and horizontal (comparing across teams within region) comparisons. Team-level analysis within each regional bar is supported by colour-coded stacks. Only those players clearing the rating >1.1 filter made the cut for an average performance above average. This can be useful for coaches who want to analyze the strength of their local talent pool, how saturated they are of a particular team, and how adept players within their region are.

R4 (CQ1): In tackling the critical question, a scatter plot that uses average_damage_per_round on the x axis and average_combat_score on the y axis was created, having the headshot_percentage as a bubble-size and the region to colour. Scatter plots are appropriate for multivariate analysis and finding correlation. Interactive filters on agent, playerName and region help the coach distill down patterns and compare performance clusters. The flexible encoding of the visual allows users to identify synergies or trade-offs between aggression (dmg), effectiveness (ACS) and precision (headshot%).

These visualisations were, together, part of a dashboard that was developed to accommodate the decision-making requirements of a coach such as Alex Kim. With this range of different visualisation types you can easily move from single-observations to entire-analysis.

3.Design

Paper Landscape Design

To brainstorm the interactive dashboard that would be developed, a prototype was sketched using the Paper Landscape method. This visualisation design is a theoretical design banging

only on theory and best practices that we learned during the module without being bound by the restrictions of Tableau or Power BI.

The design is made up of four major panels designed to have a left to right top down flow:

Panel 1: A bar chart showing the top 10 players by ACS supports ranking (Q1).

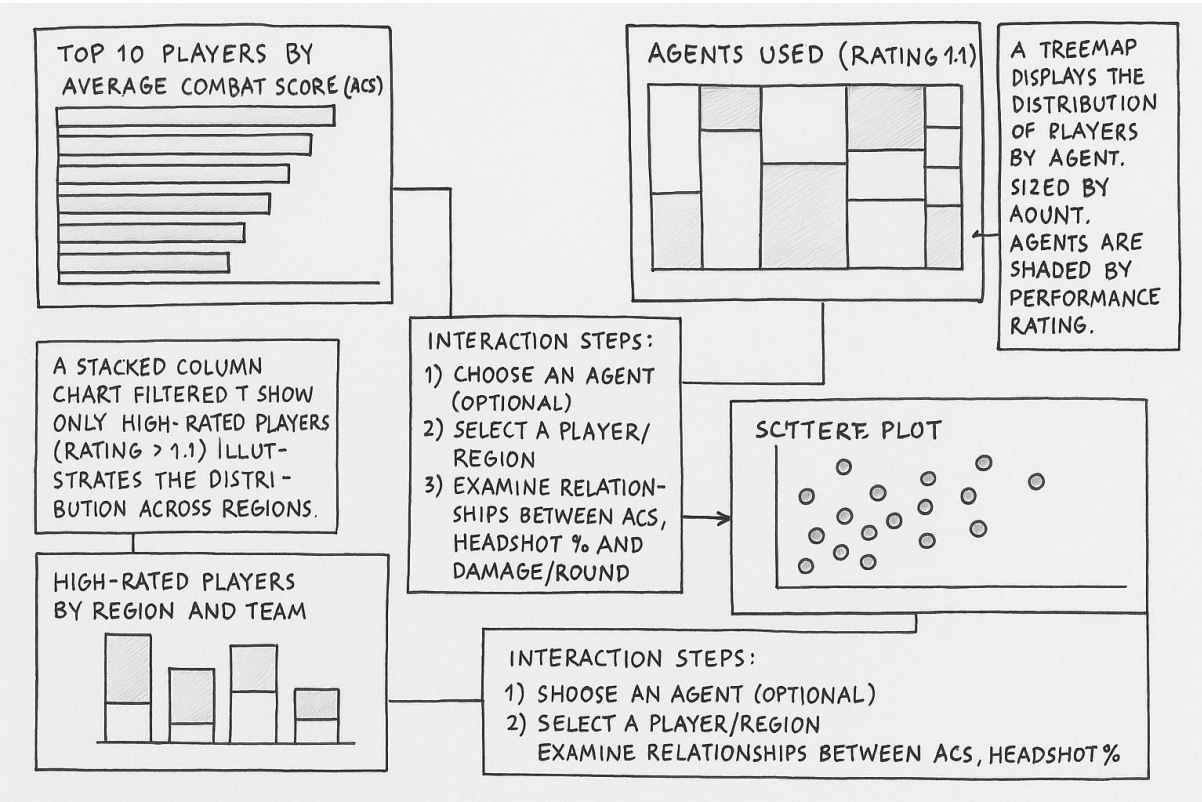
Panel 2: A treemap presentation of the agent usage distribution for high-skilled players (rating > 1.1). Box size denotes frequency; colours enclose average rating, facilitating part-to-whole judgement (Q2).

Panel 3: There are high-rated players from all regions and all teams as shown on the right side with the stacked column chart. Teams are colour-encoded in order to aid visual perception of relative strength (Q3).

4: A scatter plot showing the relationship between ACS, Headshot %, and ADR. Headshot % is encoded by circle size while region is represented by colour. This multivariate perspective addresses complex question (CQ1) and enables exploratory analysis.

Interactive features are labeled on the sketch that illustrate how users can drill-down on the insights to subset by agent, region or player. These annotations are based on Bertin's Image Theory, Mackinlay's visual encoding hierarchy, and Few's advice on when to use which chart.

Design specifications The dashboard was developed at a resolution of 1920x1080px to prevent scrollbars, all views being visible at all times. Filters are organized in the sidebar to reduce visual distractions, in line with common dashboard layout principles.



Note: The Paper Landscape sketch was generated using ChatGPT based on design decisions and layout structure defined by the author.

4.Implementation

Tool 1: Tableau Desktop

To develop the dashboard on Tableau, questions were answered on individual worksheets and finally brought together on a dashboard.

Q1 - A horizontal bar chart was generated having playerName in Rows and average_combat_score on Columns. No sorting was used as the display list was sorted in descending order and a Top N filter was used for the top 10 ACS ranked players. team appealed to Colour to differentiate players by allegiance. Tooltips were also augmented with some additional stats like kills_per_round, clutch_success_percentage, headshot_percentage in order to get more context.

Q2: The Treemap visualisation was generated by putting agent on Label and Count(playerName) on Size. A filter was used to display only players with rating > 1.1, representing top-rated players. Colouring was such that average rating per agent was an encoded attribute. The Marks card was also utilized to place the agent's name inside each Treemap tile and font sizes were altered to make data legible.

Q3: This was done with a stacked column chart with region on the X axis, Count(playerName) for Y, and team on Colour, giving team level breakdowns by region. The view was filtered to only high-rating players (rating > 1.1). This map is especially useful to look at regional strength as well as the prevalence of powerhouse teams.

CQ1: In the case of the complex interactive question, we designed a scatter plot with average_damage_per_round and average_combat_score as X and Y axes respectively, and headshot_percentage mapped to Size and region mapped to Colour. Agent, playerName, and team were added as filtered fields on the worksheet, providing cross-view dynamic filtering. Such a configuration permitted coaches to investigate the relationships between player performance and accuracy measures and to discover role or area dependent patterns.

After prepping all sheets, a Tableau Desktop dashboard was designed in a landscape layout. Charts were put into places we felt belonged on the dashboard, filters were added as quick filters and dashboard actions. Dashboard borders, titles, and legends were beautified for more professional presentation.

Tool 2: Power BI

From Power BI, the journey started with loading the identical CSV dataset and also creating these 4 visuals.

I made the Q1 bar chart the by chart and had average_combat_score on X and playerName (at the top) as Y and put a top n filter on the players in the visual-level filters pane, would that work? team was added as Legend.

Treemap is used in the Q2, where agent was placed on Group and Count of playerName was placed on Values. A filtering rating > 1.1 was done with the filter pane.

Stacked column chart for the Q3 visual was created with region under Axis, Count(playerName) under Values and team under Legend. Filtering was done in the Filters panel.

For CQ1, a scatter graph was made with average_damage_per_round on X, average_combat_score on Y, headshot_percentage and Size, and region as Colour. Slicers for agent, team, and playerName were included to facilitate real-time filtering on all visuals.

I have customized tool tips from the tool tip fields and all visuals are placed in a single dashboard canvas and well organized. Where Tableau allowed for superior layout flexibility, Power BI was more effective in the management of slicers and filters resulting in a simplified interaction.

5.Walkthrough

Q1: Top 10 ACS

For Q1, the bar chart did a good job of identifying the top 10 performing players like TenZ (and Aspas), who actually had high ACS scores throughout the series. The table was ordered in decreasing order and for that reason it was very simple to view the top performing people. By encoding the colours by team, it enabled an additional layer of understanding, aiding coaches to link standout players to their organisations. When a user mouses over a bar, detailed information such as kills per round, kill-death ratio, and headshot percentage appear as tooltips. This contextualisation provided the coach character with better understanding of how ACS is related to a player's offensive ability and accuracy.

Q2: Most Used Agents

The Treemap in Q2 provided a clear picture of agent usage within players with a rating over 1.1. Duelists and controllers were most often chosen by high-achieving players, and all three of those agents appeared in bigger boxes, meaning that players like Jett, Raze and Omen are selected a lot by high-ELO players. The visual representation made use of size and colour to illustrate not just popularity, but also average rating. This simultaneous encoding allowed for rapid visual comparison of commonly employed agents and their relative efficacy. For a coach, it would help confirm the current team meta, or vary how roles are practiced during a scrim.

Q3: Regional Distribution

The 3rd quarter stacked column chart showed the spread of the most valuable players (quality-wise) by region and then by team. It's in regions like NA and EMEA that have had the most competitive teams, including teams like Sentinels, Fnatic, and LOUD proved to be an impressive powerhouse of talent. Our filtering with rating > 1.1 density guarantee that only top players are considered. And even the more obscure regions like APAC and South America had some interesting players -- something (APAC) -- that could force teams to look at some under-the-radar talent.

CQ1. 1: Performance Scatter in Clusters

The second clustered scatter plot allowed the coach to visualize the players being grouped into role based performance profiles. With interactive filters, the coach is able to look at the closest players in the same cluster (e.g., high-damage supports) and assess consistent performance across regions. Which furthered, deep scouting and role fitting methodologies.

6.Discussion

This work demonstrates that there are nuanced insights that interactive dashboards can reveal, particularly in areas characterised by competition such as esports. The coordinated visualisation methods provided comparative, categorical, and correlational analysis which were important to meet the coach persona's tactical requirements. The visualizations individually offered their own lens, and together, the dash lent a layered decision framework – from pinpoint player performance to examination of country mappings and regional landscapes.

Take the user by the hand… major update: May 11 2014 The very first thing I've learned while doing this project: when it comes to teaching, user-friendly design is a must. Alex Kim, the persona, needed clear comparisons and trends in the numbers. This resulted in favouring visualisations that facilitated ranking (bar chart), distribution (treemap and stacked column) and multivariate exploration (scatter plot). These design choices were validated as we interactively looked through data, using filters and visual cues to test on-the-fly hypotheses (e.g., whether aggressive duelists like Jett were showing up disproportionately often in top-performing teams across every region).

There were some technical difficulties during the data processing. The agent field needed to be normalised as several names were seperated by commas in a single cell. This was solved using Python by duplicating rows to generate the vct_players_split_by_agent. csv file. Further cleaning steps involved the deletion of unfinished rows and the normalisation of teams and

region names. A second consideration was to balance layout structure between two platforms. The aesthetic features, such as color, font, and level of detail on the layout, were more optimizable in Tableau, but the slicers and interactive filters were more seamlessly integrated in Power BI.

One important insight we could get from the dashboard are the clustered roles of the players by area. Duelist roles — which had high ACS and headshot percentages — were more prevalent in North America and EMEA, whereas support or control agents were more popular in APAC and South America. This role dichotomy has real implications for coaching, specifically regarding enemy prep and scrim scheduling.

In relation to tool performance, Tableau's dashboard actions enabled smoother building of interactivity between visualisations. Yet the clean aesthetics and well-organized data model in Power BI made it easier to keep the user experience clean while adjusting datasets. Parameterized filtering was a common emerge feature in both the tools allowing for focused analysis.

For the future, the dashboard could be improved by incorporating clustering algorithms to identify latent player performance profiles. Predicting the trends over matches, or conducting a time-series analysis of agent popularity, could also bring a longitudinal perspective. Adding player sentiment or behaviour metrics (e.g., decision latency, clutch behaviour) could further enhance tactical use-cases for pro coach staff.

In conclusion, in addition to showcasing the analytical potential of visual tools in esports, this project also emphasized the benefits of contextualised and user-centred design in performance analytics.

The dashboard was designed following HCI and UCD principles to improve usability and enable sensemaking. Shneiderman's mantra — overview first, zoom and filter, then details-on-demand — was a guiding principle. They start with high-level rank and distribution charts then downward through interactive filters to end with a multivariate scatterplot.

Further, the design was guided by Nielsen's heuristics for usability:

Colour mapping via charts (e.g. team colours) maintained consistency and standards.

Recognition rather than recall was facilitated by labelled axes, legends and visible filters.

User control and freedom were provided through the interactive filters and reset options, which let the coach test multiple hypotheses unconstrained.

This strategic framework reinforced the relevancy of the dashboard with decision-making processes and increased its value for the targeted persona.

7.Conclusion

The dashboard effectively met the needs of the coach persona with a multi-chart structure for 3 pre-structured and 1 ad hoc question. Each visualization was designed to serve a key decision-support role: to highlight 'single' stars, to visualize meta-popular agents, and to compareregional & team performance. The mutuality of visual analytics tools allowed layered understanding from statistical minutiae of the micro to broad long-term trends of the macro, which translated to effective scouting, and preparation, and could also inform team strategy adjustments.

We intentionally kept the visual design simple and interactive. Bar charts and stacked columns brought categorical comparisons home to roost, or treemaps showing ratios in use of agents. The bubble chart allowed for on-the-fly analysis with filters in several dimensions. They empowered the user persona, Alex Kim, to make decisions about data without the flood-and-drought cycle of 'death by waterfall data' or no analysis at all.

Tableau and Power BI utilization reinforced technical skills in visual encoding, Top N filter application, dynamic dashboards, and parameterized design. Tableau was a rich playground of layout accuracy and cross visual filter action, whereas Power BI allowed fast dashboards to be created with responsive design dashboards and an efficient way of dealing with slicers. The duality of working on the two platforms underscored the need of a proper fit of visual solutions, so that they could harness tool strengths, yet grounded in analytical consistency.

Further, the project showed that if human-centred design methods in general, and the user persona more specifically, are used in the process of designing visualisation tools, data

visualisation can be more relevant and useful. In focusing the dashboard around what would be useful to a professional esports coach, we ended up creating a system that not just reported data effectively, but gave a nod to real-world ways of making decisions. This design driven approach was to ensure that visualizations were not just data displays – they were interactive agents-for-insight.

In the end, the project made it possible to demonstrate how game-like, well-scoped interactive visualization project can provide a competitive value in business domains such as esports. It would be interesting to develop future improvement as it would further increase the analytics capabilities, however, as it is, the dashboard gives an immediate strategic value to the user.

8. References

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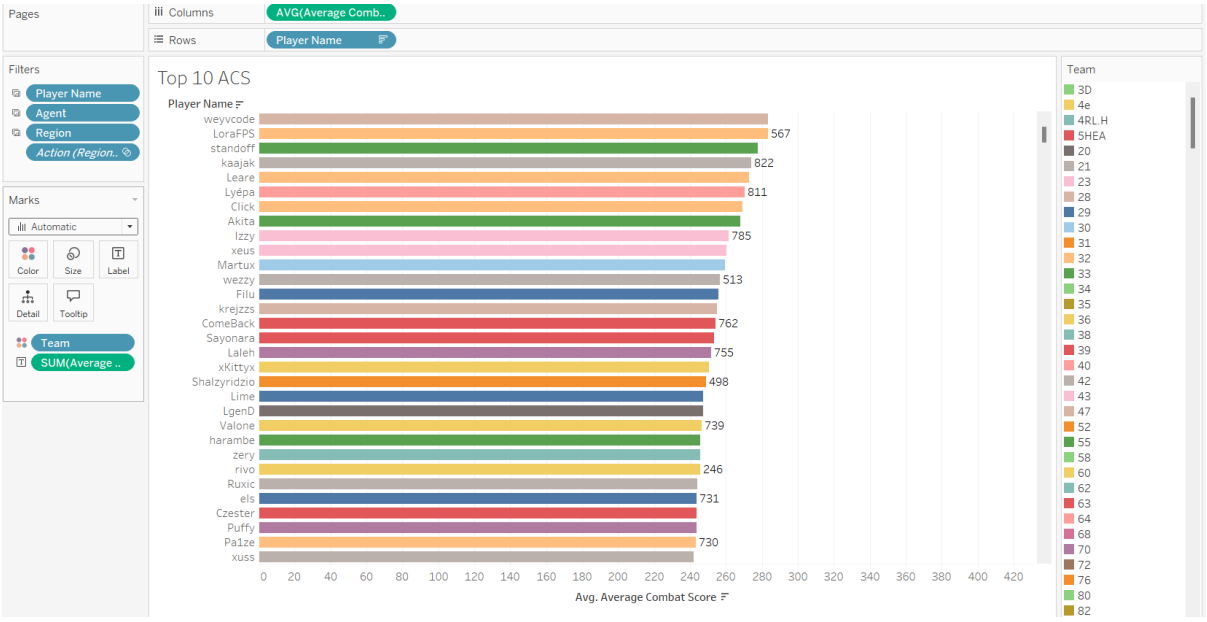
9.Appendix

SQ1:



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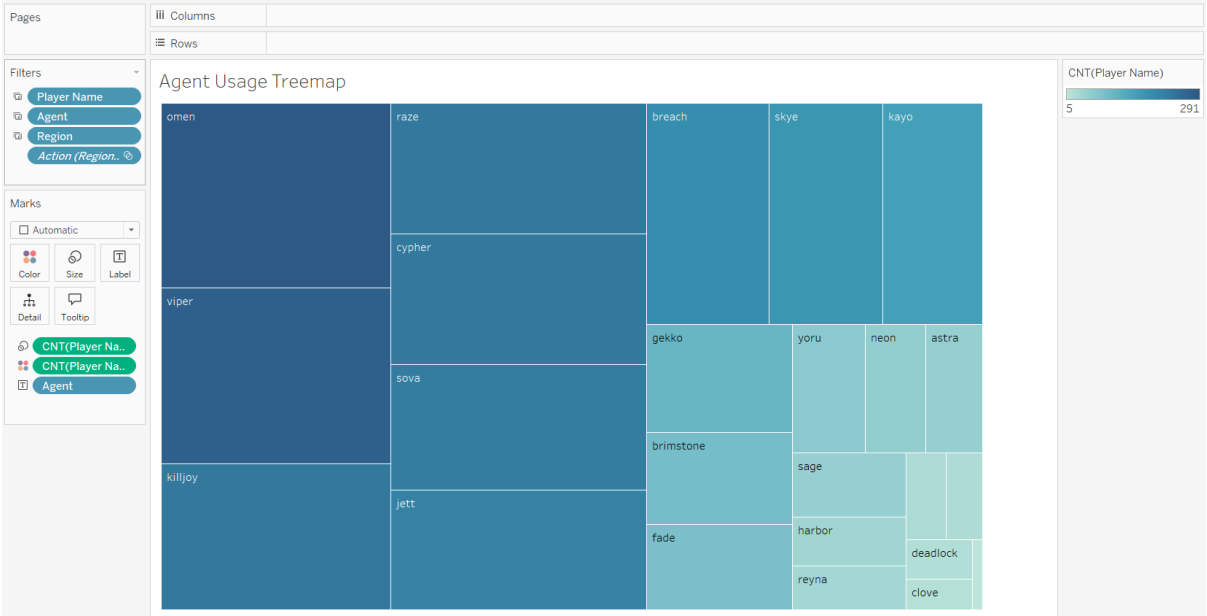
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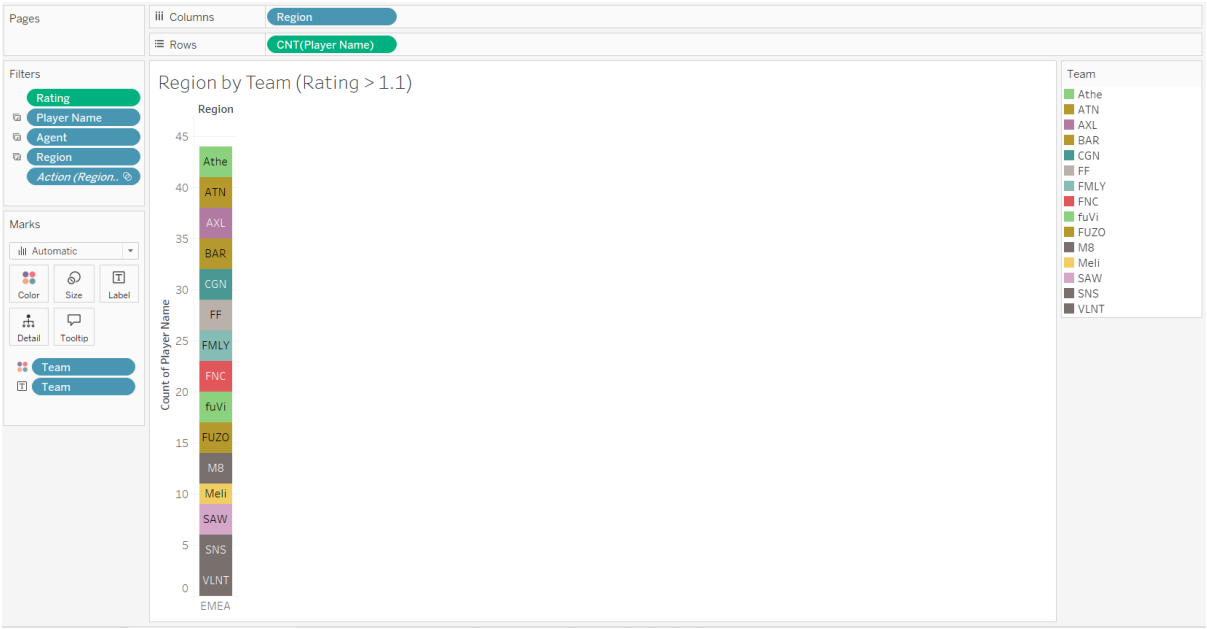
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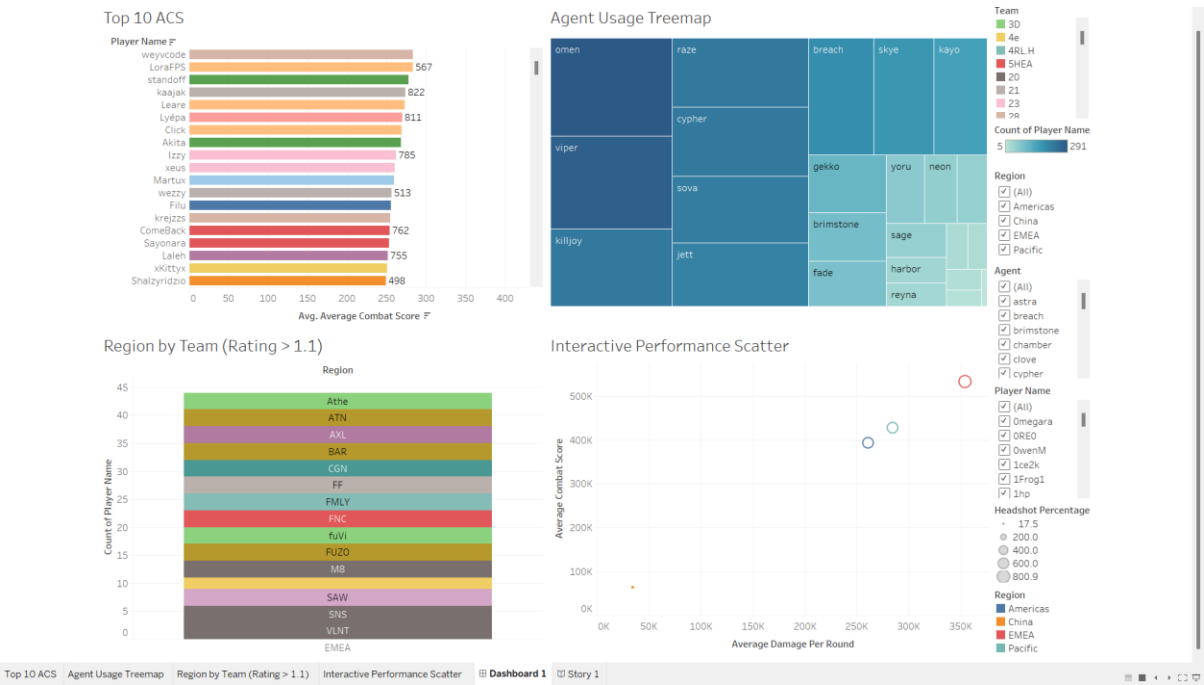
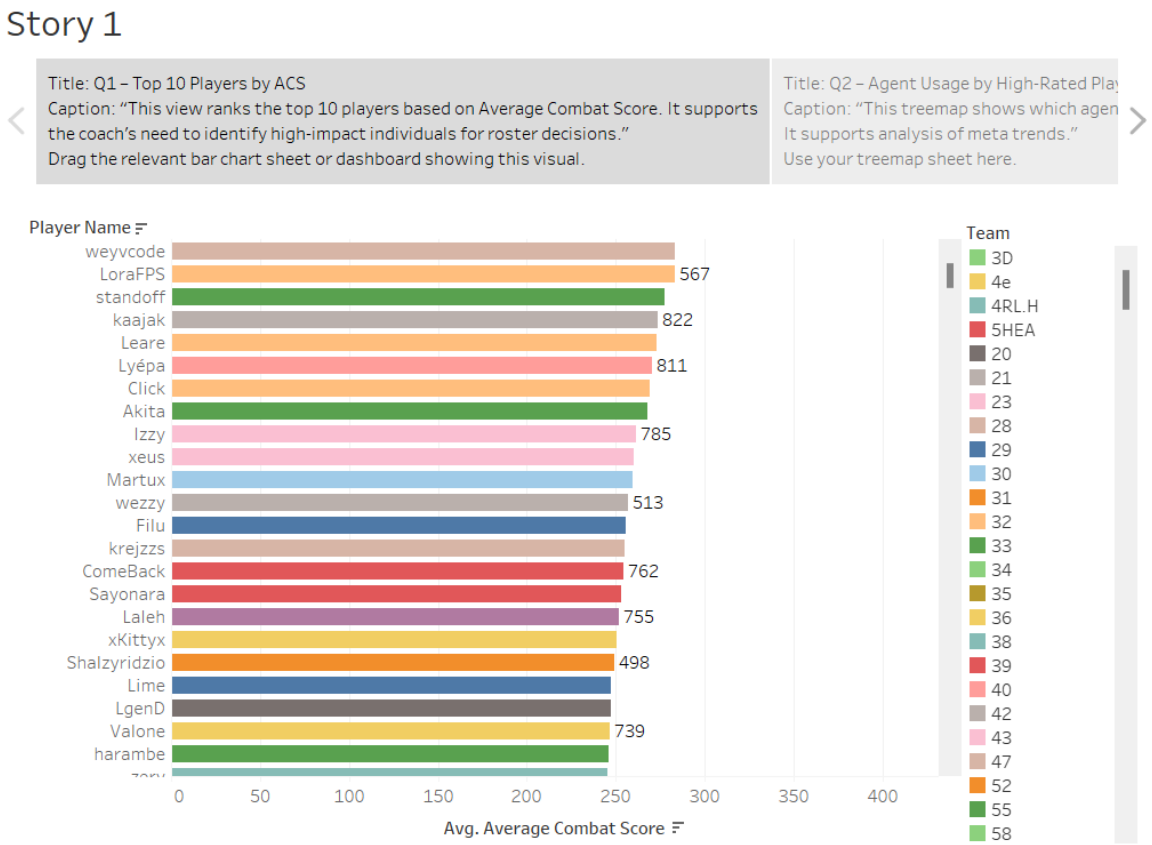
CQ1:

Tableau Dashboard:



Story:


Story 1
Title: Q1 – Top 10 Players by ACS
Caption: "This view ranks the top 10 players based on Average Combat Score. It supports the coach's need to identify high-impact individuals for roster decisions."
Drag the relevant bar chart sheet or dashboard showing this visual.
Title: Q2 – Agent Usage by High-Rated Play
Caption: "This treemap shows which agen
It supports analysis of meta trends."
Use your treemap sheet here.
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LoraFPS
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Akita
Izzy
xeus
Martux
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Filu
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ComeBack
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Story 1

Average Combat Score. It supports
roster decisions."
g this visual.

Title: Q2 – Agent Usage by High-Rated Players
Caption: "This treemap shows which agents are most used by players with a rating > 1.1.
It supports analysis of meta trends."
Use your treemap sheet here.

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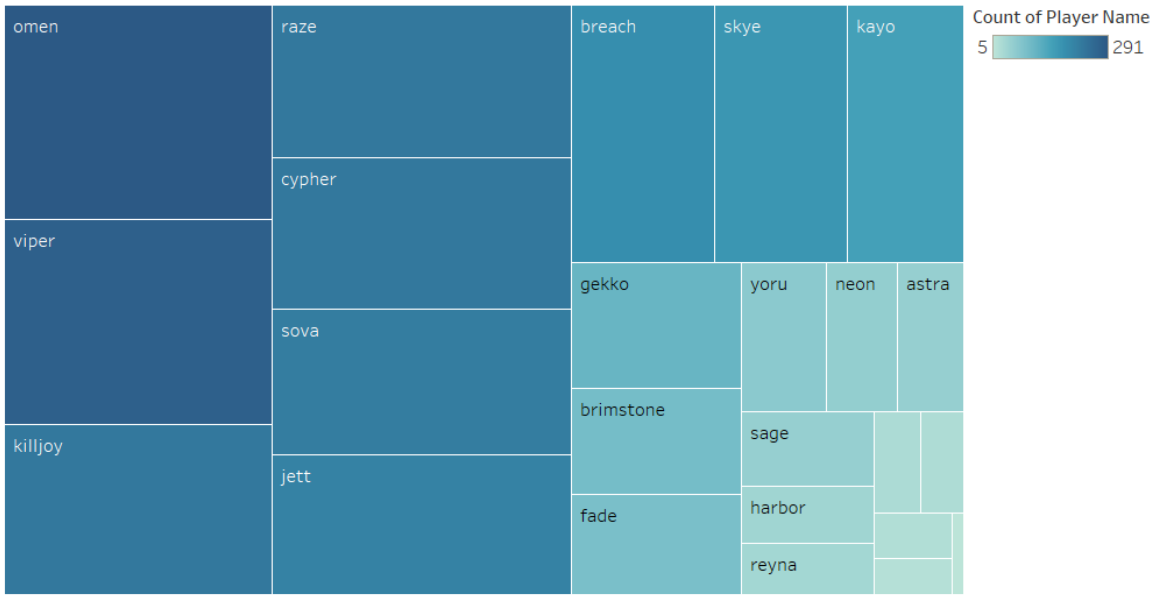

Story 1

used by players with a rating > 1.1.

Title: Q3 – Distribution by Region and Team
Caption: "This stacked column chart shows how high-rated players are distributed across regions and their teams. It highlights inter-regional strengths."
Add the stacked column chart sheet or dashboard.

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Capti
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Story 1

v high-rated players are distributed
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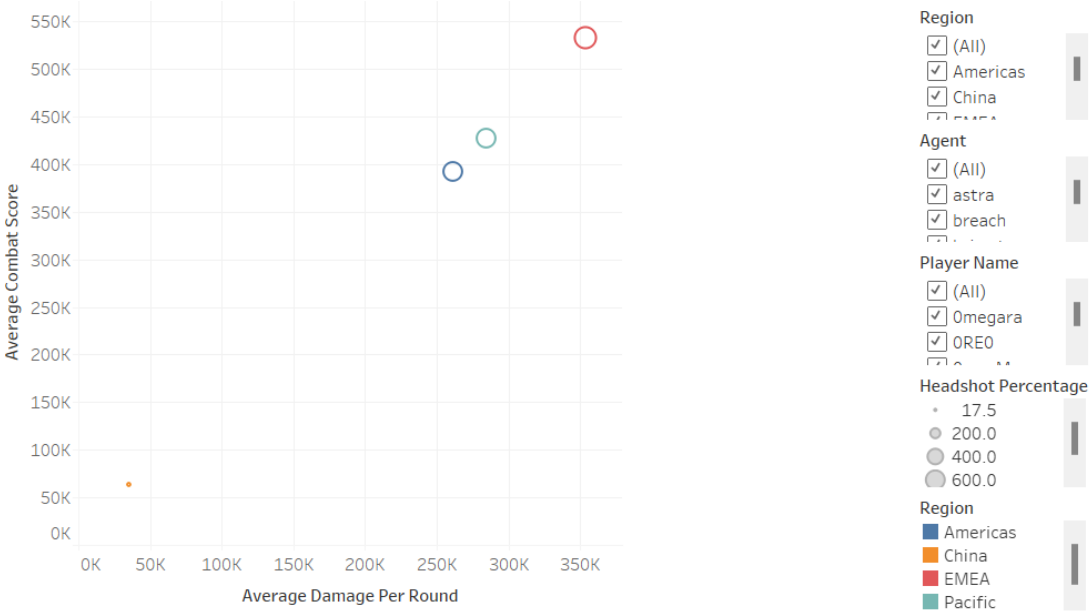
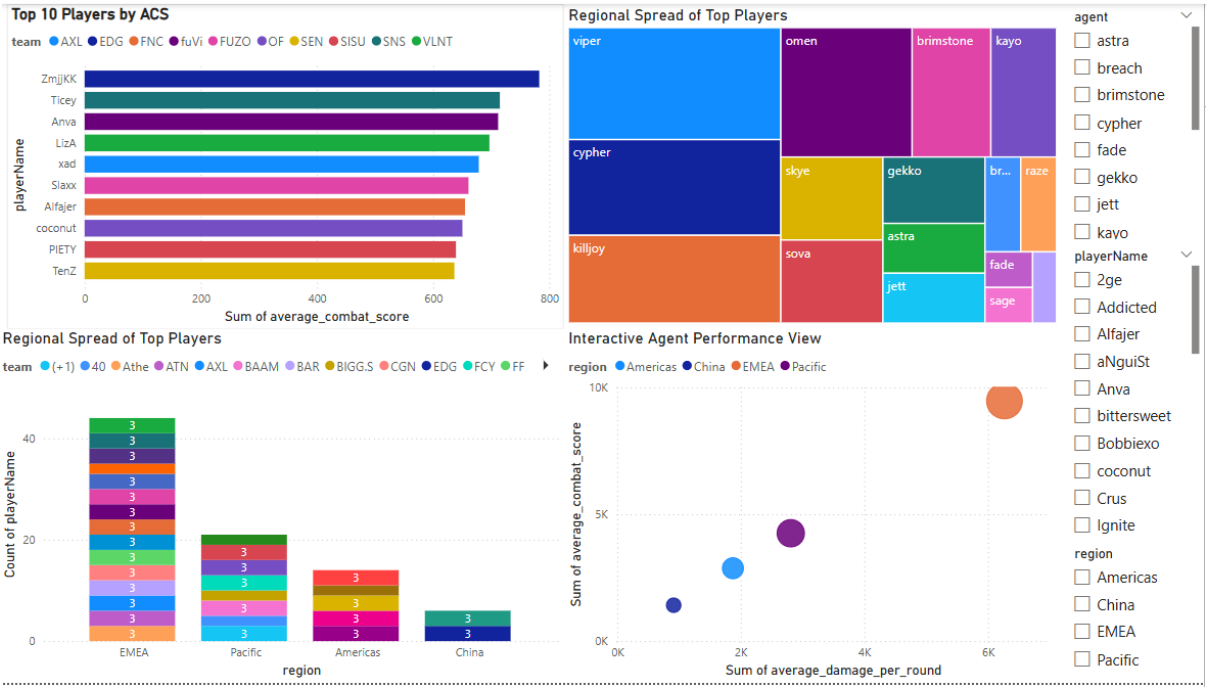
Title: CQ1 – Performance Comparison (Scatter Plot)
Caption: "This scatter plot enables multivariate comparison using ACS, ADR, and Headshot %. Filters allow role- and region-specific insight."
Use the scatter plot worksheet.



Power BI: Dashboard